

Social Media REST API – Assignment Day 4

1. Description

This RESTful API enables core features of a **Social Media Web Application**, separating concerns into frontend, backend, and data storage layers. It provides the following functionality:

- User registration and login
- 2-Factor Authentication via email
- Session management with JWT tokens
- Friend requests (send/accept/reject)
- Creating, liking, and listing posts

2. Authentication

- **Signup** requires email verification for 2FA.
- **Login** generates a JWT token.
- The token is used for authorizing future requests via the Authorization Bearer<token> header.
- JWT is preferably stored in **HttpOnly secure cookies, not in localStorage** (to avoid XSS attacks).
- Tokens include user data and an expiration timestamp.
- All protected routes check JWT validity and role-based permissions.

Note: If the backend is not integrated, login can still simulate token issuance, but full functionality (DB sync, friend requests, etc.) requires backend implementation.

3. Base URL

https://socialmedia

4. API Indexing Table

#	Feature	Method	Endpoint	Description
1	User Sign-up	POST	/signup	Register new user
2	User Login	POST	/login	Authenticate user, return JWT
3	Send Friend Request	POST	/friend-request	Send request to another user
4	Respond to Friend Request	POST	/friend-request/respond	Accept or reject incoming friend request
5	Create Post	POST	/posts	Create a new post
6	Like Post	POST	/posts/{post_id}/like	Like a post once
7	List Posts	GET	/posts	View posts from self and friends

5. Request Schema Summary

Field	Type	Notes
Username	string	Unique, 3–30 characters, no special characters
Email	string	Must be valid format
Password	string	Min 8 chars, includes upper, lower, number, special char
fuser_id	string	Friend's user ID
content	string	Post content (max 256 characters)
post_id	string	Post identifier

Field	Type	Notes
request	string	"accept" or "reject"

6. Credit Points (Functional Highlights)

- JWT-based secure sessions
- 2FA during signup
- Path and query parameters used correctly
- Rate limiting and error handling built in
- Content validation and authorization middleware
- Secure HTTPS usage recommended
- Roles and permissions encouraged in backend design

7. Request / Response + Response Codes

User Sign-Up

POST /signup
Content-Type: application/json

Request

```
{
  "username": "Strot Jain",
  "password": "Strot@1234",
  "email": "strot.jain@syvora.com"
}
```

Response

```
{
  "message": "User registered successfully",
}
```

```
"user_id": "123456"
}
```

Response Codes

- 200 OK
 - 400 Bad Request
 - 409 Conflict
-

User Login

```
POST /login
Content-Type: application/json
```

Request

```
{
  "email": "strot.jain@syvora.com",
  "password": "Strot@1234"
}
```

Response

```
{
  "message": "User Logged In",
  "token": "JWT-token",
  "user_id": "123456"
}
```

Response Codes

- 200 OK
 - 401 Unauthorized
-

Send Friend Request

Request

```
{
  "fuser_id": "654321"
}
```

Response

```
{
  "message": "Friend-Request sent",
  "fuser_id": "654321"
}
```

Response Codes

- 202 Accepted
- 410 Gone
- 429 Too Many Requests

Respond to Friend Request

POST /friend-request/respond
Authorization: Bearer <JWT-token>

Request

```
{
  "user_id": "789123",
  "request": "accept" / "reject"
}
```

Response

```
{
  "message": "Friend request accepted",
}
```

```
"user_id": "789123"
}
```

Response Codes

- 200 OK
 - 400 Bad Request
-

Create a Post

```
POST /posts
Authorization: Bearer <JWT-token>
```

Request

```
{
  "content": "Hello world! This is my first post."
}
```

Response

```
{
  "message": "Post created",
  "post_id": "post001"
}
```

Response Codes

- 201 Created
 - 400 Bad Request
-

Like a Post

```
POST /posts/{post_id}/like
Authorization: Bearer <JWT-token>
```

Response

```
{
  "message": "Liked"
}
```

Response Codes

- 200 OK
 - 409 Conflict (already liked)
 - 404 Not Found
-

List Posts

```
GET /posts
Authorization: Bearer <JWT-token>
```

Response

```
[
  {
    "post_id": "post001",
    "content": "Hello world!",
    "author": "Strot Jain",
    "likes": 10
  }
]
```

Response Codes

- 200 OK
 - 401 Unauthorized
-